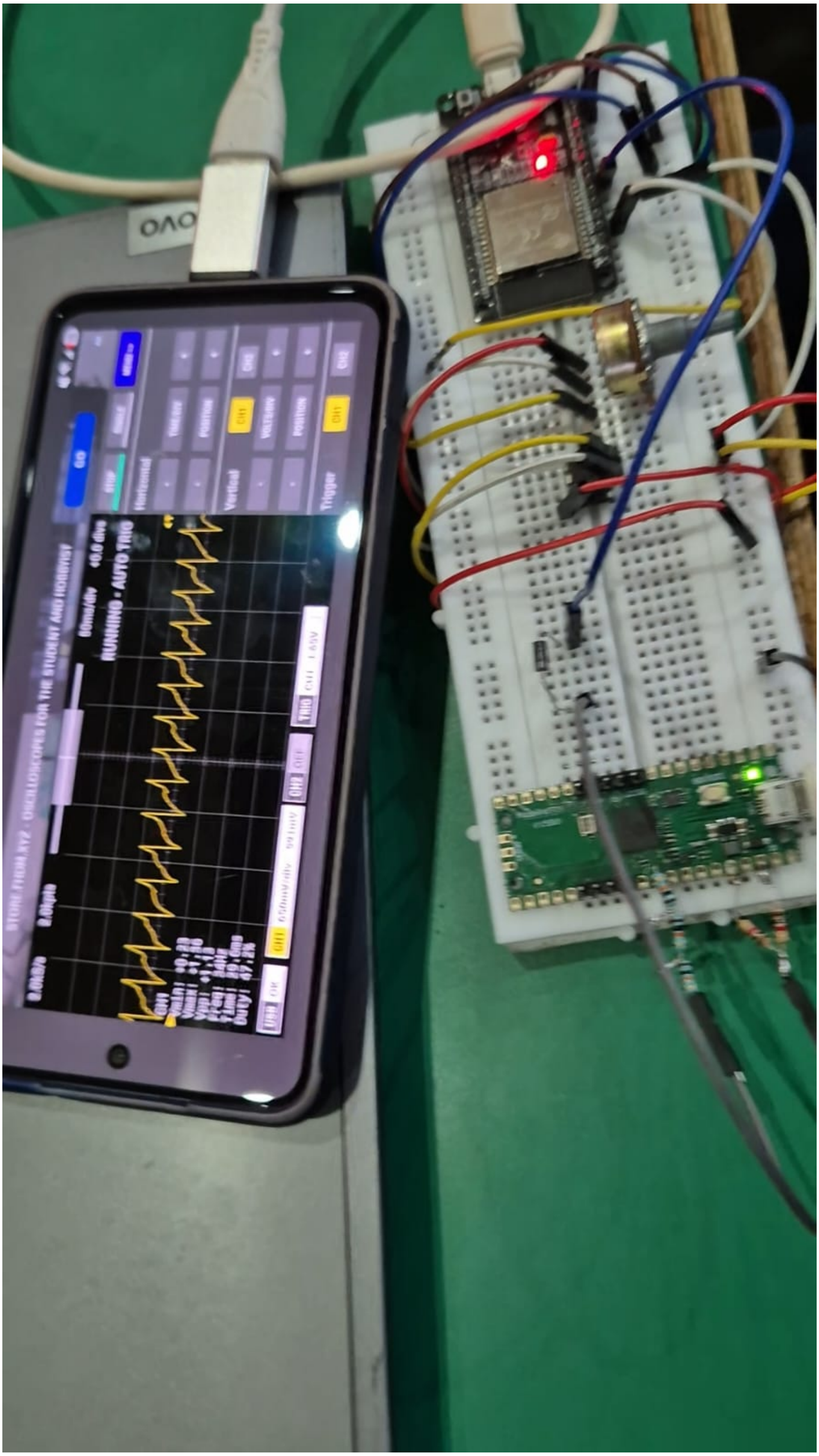
# Pocket Lab

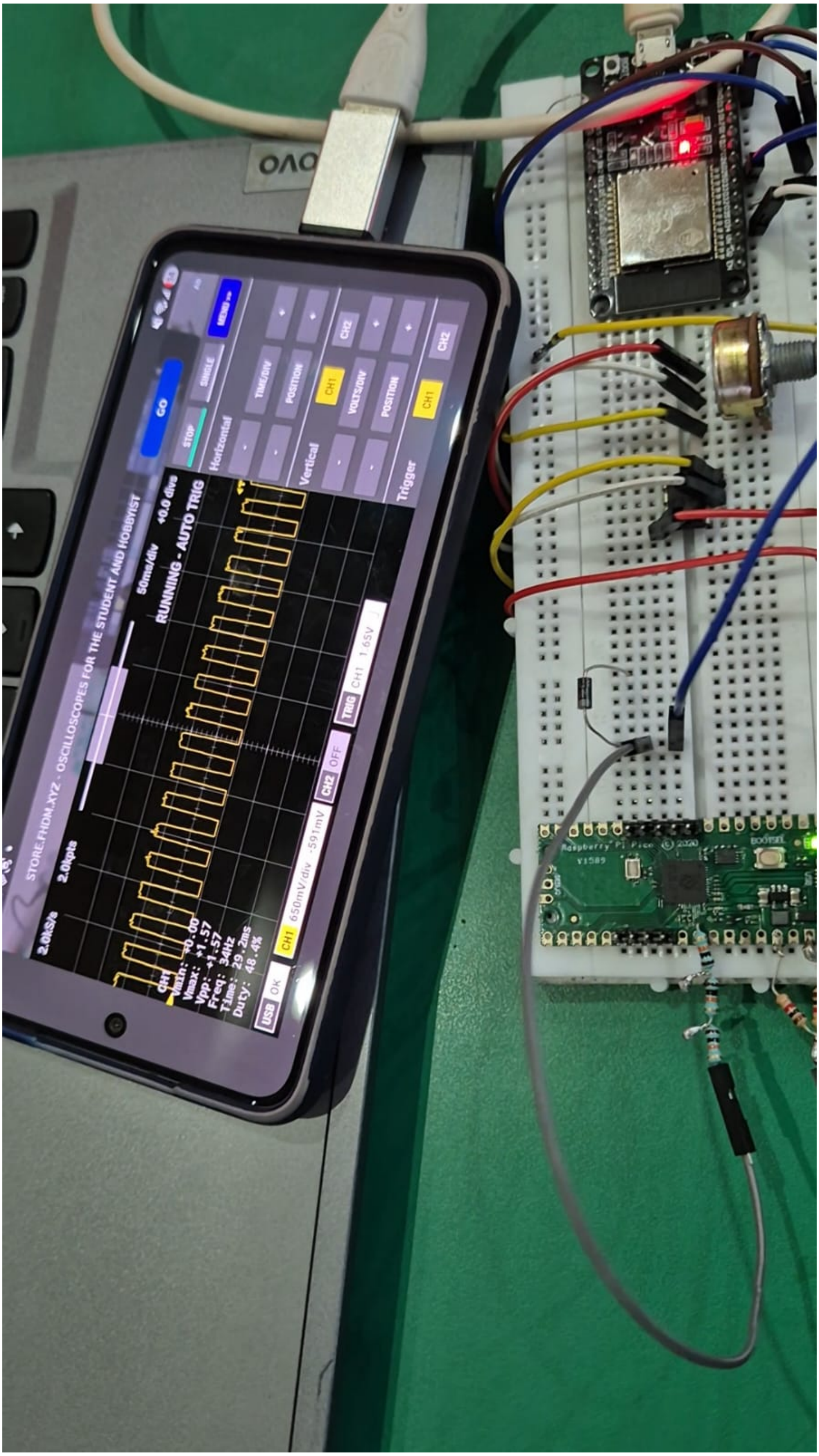
## Hardware Circuitry

Build photos, test captures, and component reference

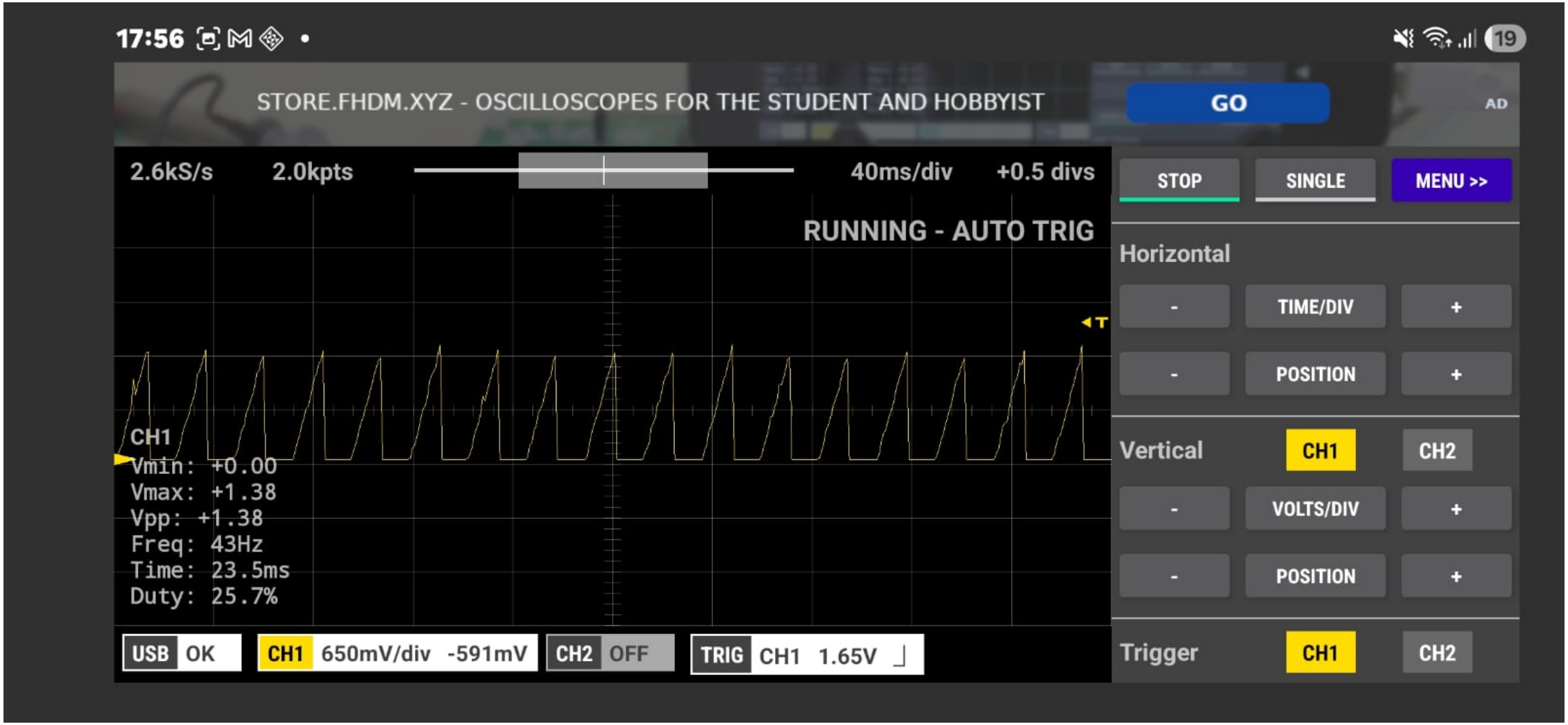
*IICH Hackathon, Delhi Technological University, 2025*



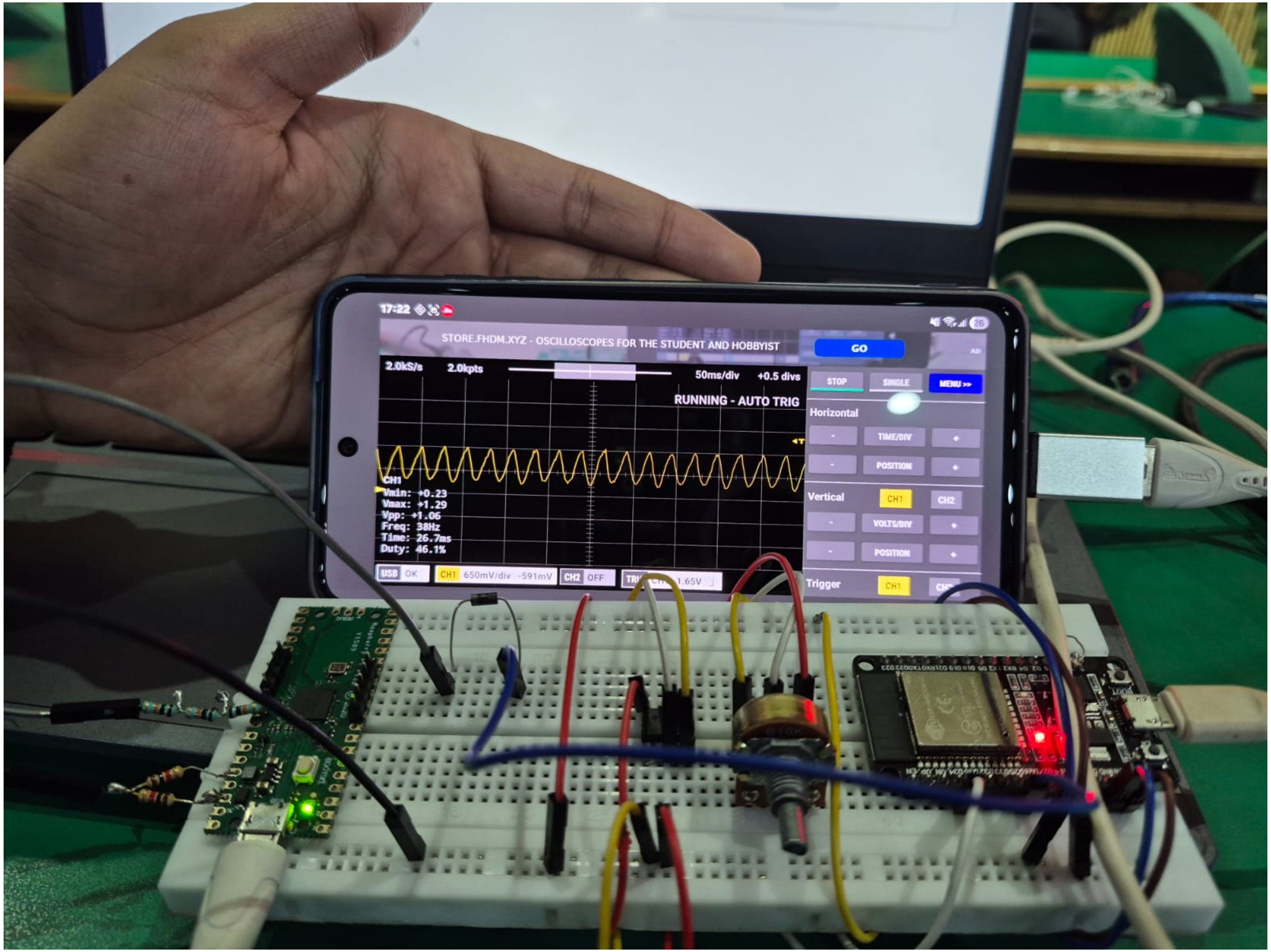
The complete Pocket Lab setup: the Android phone (running the Scoppy app) held alongside the full breadboard circuit — ESP32 function generator, Raspberry Pi Pico running Scoppy, and the signal-conditioning components, all connected and live.



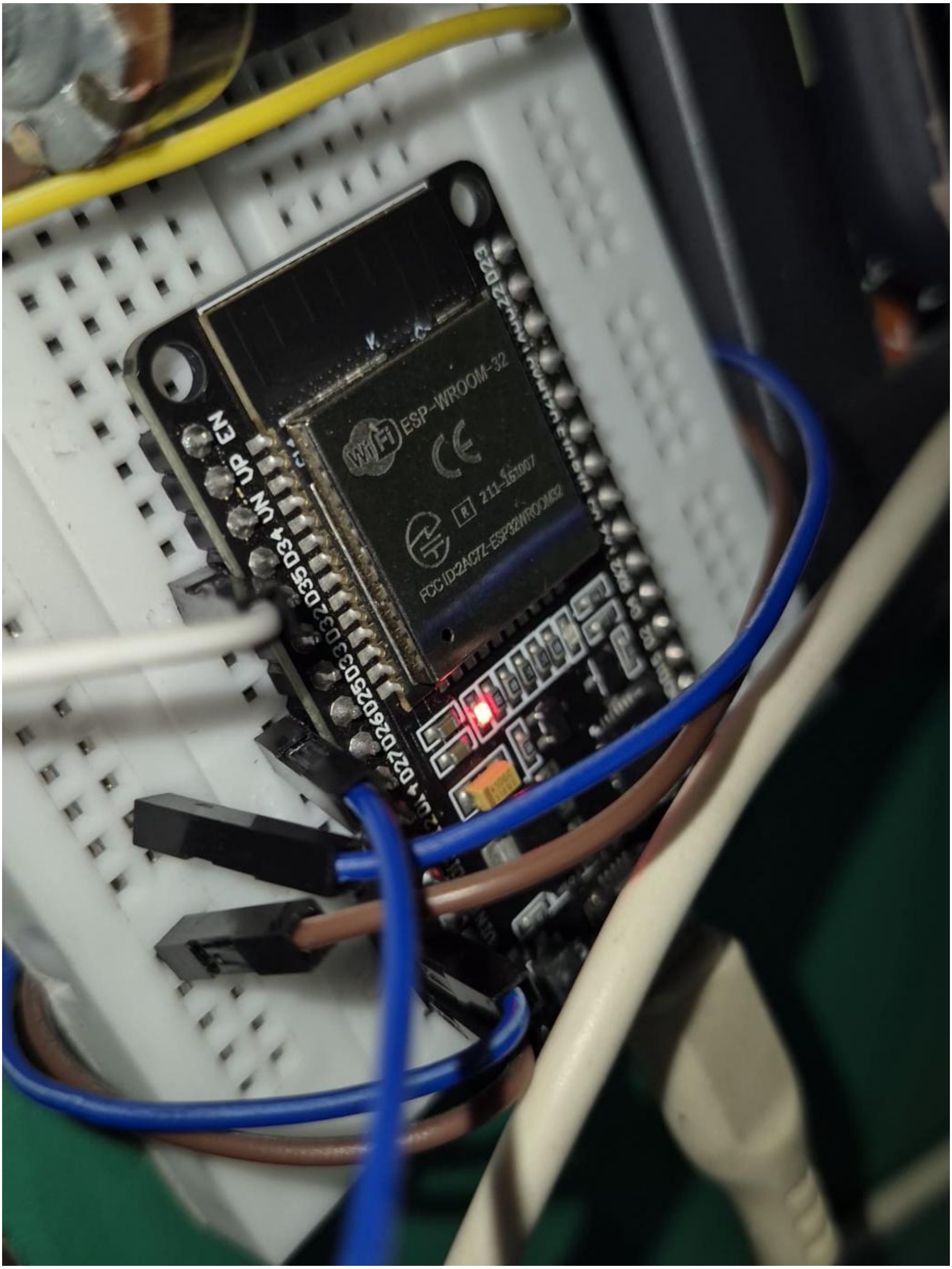

The breadboard circuit captured mid-test, with the phone screen showing the Scoppy oscilloscope interface actively reading a sawtooth waveform from the circuit.




A closer look at the Scoppy app screen: a sawtooth waveform captured on Channel 1, running at 43Hz with a peak-to-peak voltage of 1.38V.



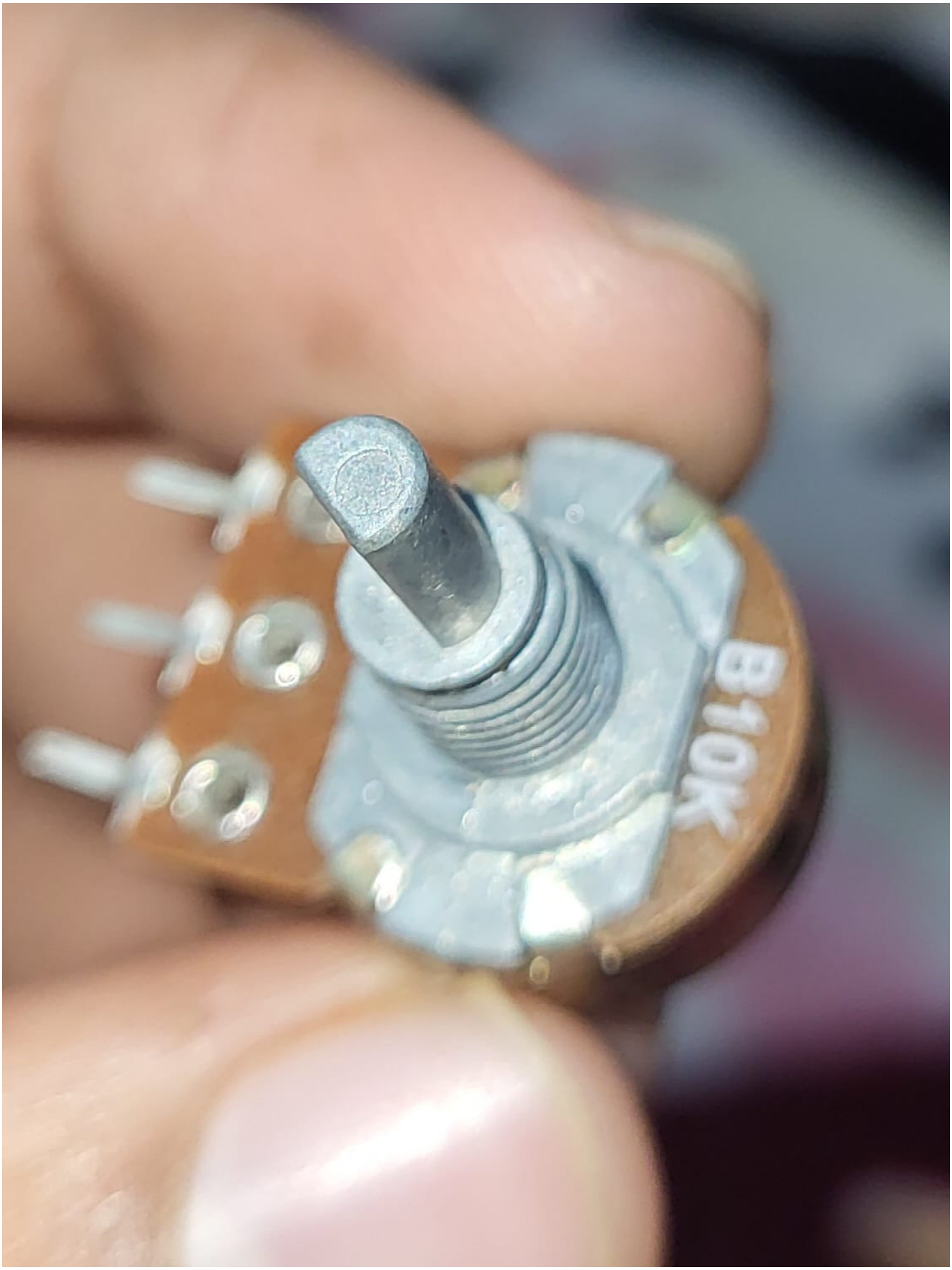

The setup again, this time displaying a sine waveform on the Scoppy oscilloscope screen — generated by the ESP32 and measured through the RP2040 + phone combination.




A close-up of the ESP32 (WROOM-32) DevKit on the breadboard — the microcontroller responsible for generating all four function-generator waveforms (sine, square, triangular, sawtooth).


The NE555 timer IC — used in the earlier, pre-ESP32 attempt at sawtooth wave generation via a relaxation-oscillator configuration.


A CD4017BE decade counter IC, used during circuit experimentation for waveform and signal-sequencing logic on the breadboard.




A B10K potentiometer, used for manual adjustment of signal parameters (such as amplitude or bias) within the test and function-generator circuits.

Team Pocket Lab at work during the IICH Hackathon (DTU, 2025) — debugging the circuit, cross-checking waveform readings, and documenting notes as the build came together.